



NIŞANTAŞI
ÜNİVERSİTESİ

NIŞANTAŞI UNIVERSITY

Faculty of Economics, Administrative and Social Sciences

Management Information Systems

**Employee Salary Tracking Program - Technical Architecture
Report**

May – 2026

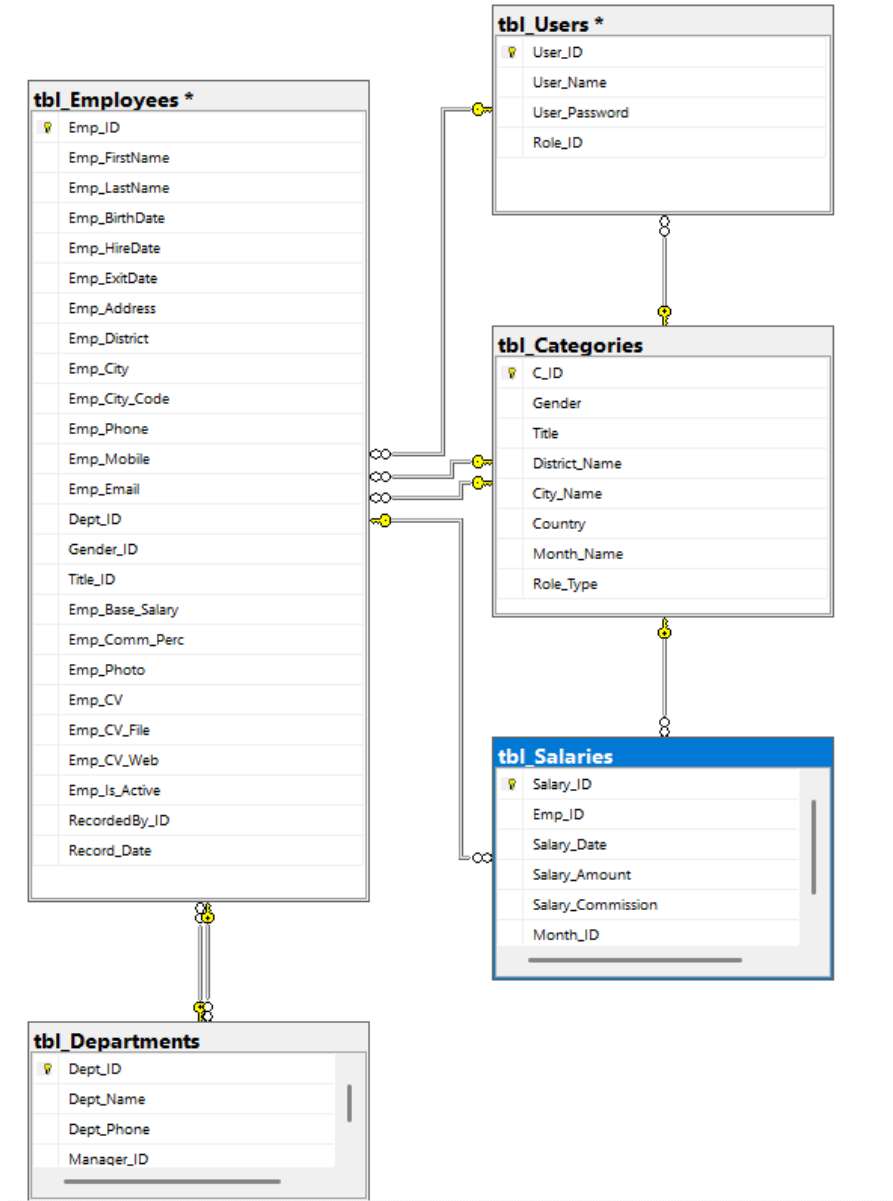
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Department: Management Information Systems

Database Conceptual Model



Database Logical Model

1. USER ENTITY

Özellik (Attribute)	Veri Tipi (Data Type)	Anahtar (Key)
User_ID	Sayı (Integer)	Primary Key (PK)
User_Name	Metin (String)	
User_Password	Metin (String)	

Role_ID	Sayı (Integer)	Foreign Key (FK)
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2. CATEGORY ENTITY (CATEGORY)

Özellik (Attribute)	Veri Tipi (Data Type)	Anahtar (Key)
C_ID	Sayı (Integer)	Primary Key (PK)
Gender	Metin (String)	
Title	Metin (String)	
District_Name	Metin (String)	
City_Name	Metin (String)	
Country	Metin (String)	
Month_Name	Metin (String)	
Role_Type	Metin (String)	

3. DEPARTMENT ENTITY (DEPARTMENT)

Özellik (Attribute)	Veri Tipi (Data Type)	Anahtar (Key)
Dept_ID	Sayı (Integer)	Primary Key (PK)
Dept_Name	Metin (String)	
Dept_Phone	Metin (String)	
Manager_ID	Sayı (Integer)	Foreign Key (FK)

4. SALARY ENTITY (SALARY)

Özellik (Attribute)	Veri Tipi (Data Type)	Anahtar (Key)
Salary_ID	Sayı (Integer)	Primary Key (PK)

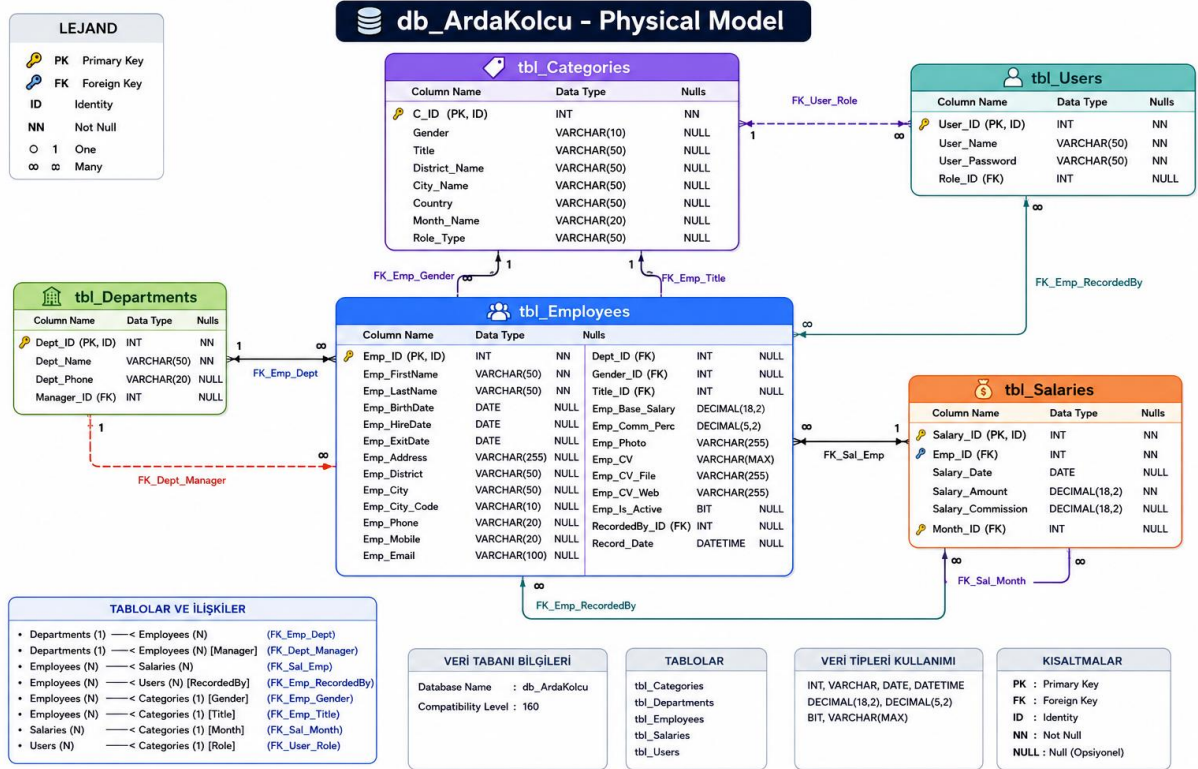
Emp_ID	Sayı (Integer)	Foreign Key (FK)
Salary_Date	Tarih (Date)	
Salary_Amount	Para / Kùsuratlı Sayı (Decimal)	
Salary_Commission	Para / Kùsuratlı Sayı (Decimal)	
Month_ID	Sayı (Integer)	Foreign Key (FK)

5. EMPLOYEE ENTITY (EMPLOYEE)

Özellik (Attribute)	Veri Tipi (Data Type)	Anahtar (Key)
Emp_ID	Sayı (Integer)	Primary Key (PK)
Emp_FirstName	Metin (String)	
Emp_LastName	Metin (String)	
Emp_BirthDate	Tarih (Date)	
Emp_HireDate	Tarih (Date)	
Emp_ExitDate	Tarih (Date)	
Emp_Base_Salary	Para / Kùsuratlı Sayı (Decimal)	
Emp_Comm_Perc	Para / Kùsuratlı Sayı (Decimal)	
Emp_Is_Active	Mantıksal (Boolean)	
Record_Date	Tarih ve Saat (DateTime)	
<i>(Adres, Tel, CV vb. diğér tüm alanlar)</i>	Metin (String)	
Dept_ID	Sayı (Integer)	Foreign Key (FK)

Gender_ID	Sayı (Integer)	Foreign Key (FK)
Title_ID	Sayı (Integer)	Foreign Key (FK)
RecordedBy_ID	Sayı (Integer)	Foreign Key (FK)

Database Physical Model



CREATE DATABASE db_ArdaKolcu

GO

ALTER DATABASE db_ArdaKolcu SET COMPATIBILITY_LEVEL = 160

GO

USE db_ArdaKolcu

GO

```
CREATE TABLE tbl_Users (
    User_ID INT PRIMARY KEY IDENTITY(1,1),
    User_Name VARCHAR(50) NOT NULL,
    User_Password VARCHAR(50) NOT NULL,
    Role_ID INT)
GO
```

```
CREATE TABLE tbl_Categories (
    C_ID INT PRIMARY KEY IDENTITY(1,1),
    Gender VARCHAR(10),
    Title VARCHAR(50),
    District_Name VARCHAR(50),
    City_Name VARCHAR(50),
    Country VARCHAR(50),
    Month_Name VARCHAR(20),
```

```
Role_Type VARCHAR(50))
GO
```

```
CREATE TABLE tbl_Departments (
  Dept_ID INT PRIMARY KEY IDENTITY(1,1),
  Dept_Name VARCHAR(50) NOT NULL,
  Dept_Phone VARCHAR(20),
  Manager_ID INT)
GO
```

```
CREATE TABLE tbl_Employees (
  Emp_ID INT PRIMARY KEY IDENTITY(1,1),
  Emp_FirstName VARCHAR(50) NOT NULL,
  Emp_LastName VARCHAR(50) NOT NULL,
  Emp_BirthDate DATE,
  Emp_HireDate DATE,
  Emp_ExitDate DATE,
  Emp_Address VARCHAR(255),
  Emp_District VARCHAR(50),
  Emp_City VARCHAR(50),
  Emp_City_Code VARCHAR(10),
  Emp_Phone VARCHAR(20),
  Emp_Mobile VARCHAR(20),
  Emp_Email VARCHAR(100),
  Dept_ID INT,
  Gender_ID INT,
  Title_ID INT,
  Emp_Base_Salary DECIMAL(18,2),
  Emp_Comm_Perc DECIMAL(5,2),
  Emp_Photo VARCHAR(255),
  Emp_CV VARCHAR(MAX),
  Emp_CV_File VARCHAR(255),
  Emp_CV_Web VARCHAR(255),
  Emp_Is_Active BIT,
  RecordedBy_ID INT,
  Record_Date DATETIME)
GO
```

```
CREATE TABLE tbl_Salaries (
  Salary_ID INT PRIMARY KEY IDENTITY(1,1),
  Emp_ID INT NOT NULL,
  Salary_Date DATE,
  Salary_Amount DECIMAL(18,2) NOT NULL,
  Salary_Commission DECIMAL(18,2),
  Month_ID INT)
GO
```

```
INSERT INTO tbl_Categories (Gender, Title, District_Name, City_Name, Country, Month_Name, Role_Type)
VALUES
('Male', 'Data Analyst', 'Sisli', 'Istanbul', 'Turkey', 'January', 'Full-Time'),
('Female', 'HR Manager', 'Kadikoy', 'Istanbul', 'Turkey', 'February', 'Contract'),
('Male', 'Software Specialist', 'Besiktas', 'Istanbul', 'Turkey', 'March', 'Part-Time'),
('Female', 'Finance Manager', 'Sariyer', 'Istanbul', 'Turkey', 'April', 'Full-Time'),
('Male', 'Intern', 'Uskudar', 'Istanbul', 'Turkey', 'May', 'Intern');
GO
```

```
INSERT INTO tbl_Departments (Dept_Name, Dept_Phone, Manager_ID)
VALUES
('Information Technology (IT)', '02125551001', 1),
('Human Resources', '02125551002', 2),
('Marketing', '02125551003', 3),
('Finance and Accounting', '02125551004', 4),
('Management', '02125551005', 5);
GO
```

```

INSERT INTO tbl_Employees (Emp_FirstName, Emp_LastName, Emp_BirthDate, Emp_HireDate, Emp_ExitDate,
Emp_Address, Emp_District, Emp_City, Emp_City_Code, Emp_Phone, Emp_Mobile, Emp_Email, Dept_ID, Gender_ID,
Title_ID, Emp_Base_Salary, Emp_Comm_Perc, Emp_Photo, Emp_CV, Emp_CV_File, Emp_CV_Web, Emp_Is_Active,
RecordedBy_ID, Record_Date)
VALUES
('Ali', 'Yilmaz', '1990-05-15', '2020-01-10', NULL, 'Ataturk Ave. No:1', 'Sisli', 'Istanbul', '34', '02125550001',
'05325550001', 'ali.yilmaz@company.com', 1, 1, 1, 35000.00, 0.00, NULL, NULL, NULL, NULL, 1, 1, GETDATE()),
('Ayse', 'Demir', '1992-08-22', '2021-03-15', NULL, 'Cumhuriyet St. No:2', 'Kadikoy', 'Istanbul', '34', '02125550002',
'05325550002', 'ayse.demir@company.com', 2, 2, 2, 40000.00, 5.00, NULL, NULL, NULL, NULL, 1, 1, GETDATE()),
('Mehmet', 'Kaya', '1988-11-30', '2019-06-01', NULL, 'Baris St. No:3', 'Besiktas', 'Istanbul', '34', '02125550003',
'05325550003', 'mehmet.kaya@company.com', 3, 1, 3, 38000.00, 10.50, NULL, NULL, NULL, NULL, 1, 2, GETDATE()),
('Fatma', 'Celik', '1995-02-14', '2022-09-01', NULL, 'Gunes Blvd. No:4', 'Sariyer', 'Istanbul', '34', '02125550004',
'05325550004', 'fatma.celik@company.com', 4, 2, 4, 45000.00, 0.00, NULL, NULL, NULL, NULL, 1, 2, GETDATE()),
('Can', 'Ozturk', '1998-07-07', '2023-01-20', NULL, 'Yildiz Ave. No:5', 'Uskudar', 'Istanbul', '34', '02125550005',
'05325550005', 'can.ozturk@company.com', 5, 1, 5, 25000.00, NULL, NULL, NULL, NULL, NULL, 1, 1, GETDATE());
GO

```

```

INSERT INTO tbl_Salaries (Emp_ID, Salary_Date, Salary_Amount, Salary_Commission, Month_ID)
VALUES
(1, '2026-05-15', 35000.00, 0.00, 5),
(2, '2026-05-15', 40000.00, 2000.00, 5),
(3, '2026-05-15', 38000.00, 3990.00, 5),
(4, '2026-05-15', 45000.00, 0.00, 5),
(5, '2026-05-15', 25000.00, 500.00, 5);
GO

```

```

INSERT INTO tbl_Users (User_Name, User_Password, Role_ID)
VALUES
('admin_arda', 'Arda1234!', 1),
('hr_merve', 'MerveHR_456', 2),
('user_ahmet', 'Ahmet789?', 3),
('user_ayse', 'AysePass321', 3),
('it_mehmet', 'ITMehmet_00', 4);
GO

```

```

ALTER TABLE tbl_Employees ADD CONSTRAINT FK_Emp_Dept FOREIGN KEY (Dept_ID) REFERENCES
tbl_Departments(Dept_ID);
ALTER TABLE tbl_Employees ADD CONSTRAINT FK_Emp_Gender FOREIGN KEY (Gender_ID) REFERENCES
tbl_Categories(C_ID);
ALTER TABLE tbl_Employees ADD CONSTRAINT FK_Emp_Title FOREIGN KEY (Title_ID) REFERENCES
tbl_Categories(C_ID);
ALTER TABLE tbl_Employees ADD CONSTRAINT FK_Emp_RecordedBy FOREIGN KEY (RecordedBy_ID) REFERENCES
tbl_Users(User_ID);
GO

```

```

ALTER TABLE tbl_Salaries ADD CONSTRAINT FK_Sal_Emp FOREIGN KEY (Emp_ID) REFERENCES
tbl_Employees(Emp_ID);
ALTER TABLE tbl_Salaries ADD CONSTRAINT FK_Sal_Month FOREIGN KEY (Month_ID) REFERENCES
tbl_Categories(C_ID);
GO

```

```

ALTER TABLE tbl_Users ADD CONSTRAINT FK_User_Role FOREIGN KEY (Role_ID) REFERENCES tbl_Categories(C_ID);
ALTER TABLE tbl_Departments ADD CONSTRAINT FK_Dept_Manager FOREIGN KEY (Manager_ID) REFERENCES
tbl_Employees(Emp_ID);
GO

```

```

SELECT * FROM tbl_Categories;
SELECT * FROM tbl_Departments;
SELECT * FROM tbl_Employees;
SELECT * FROM tbl_Salaries;
SELECT * FROM tbl_Users;
GO

```